ELEC3115
ELECTROMAGNETIC ENGINEERING

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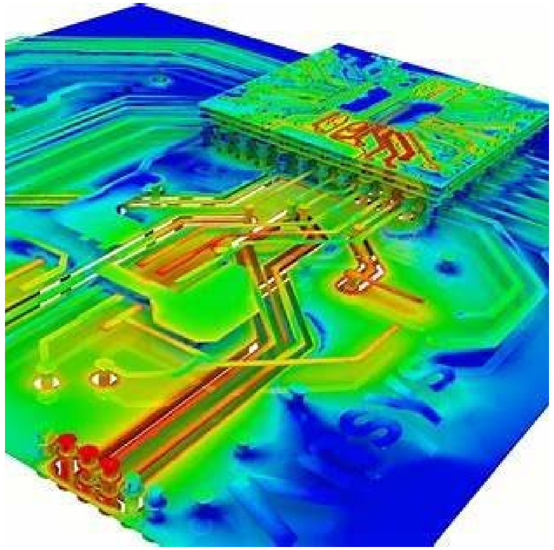


"You can't see electromagnetic fields — but they are behind all modern techs."

Why Electromagnetic Engineering Matters

Electromagnetic Fields & Waves
(the common foundation)

Power & Energy
transformers, machines, transmission lines, insulation design, field stress

Sensors & Microsystems
MEMS, inductive sensing, imaging, electrostatic actuation

Telecommunications
Antennas, wave propagation, wireless

Electronics & Devices
capacitance coupling, EMC , shielding

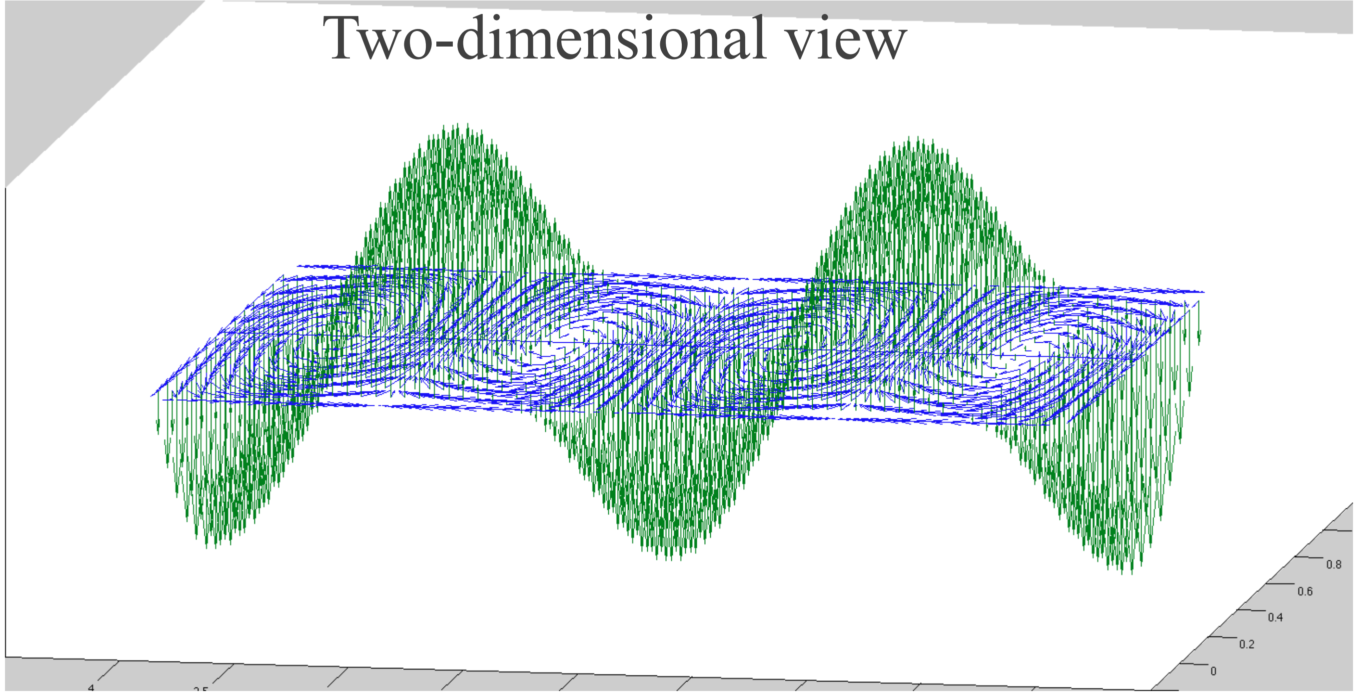
Emerging Technologies
Evs, super capacitors, radar, wireless power,5G/6G

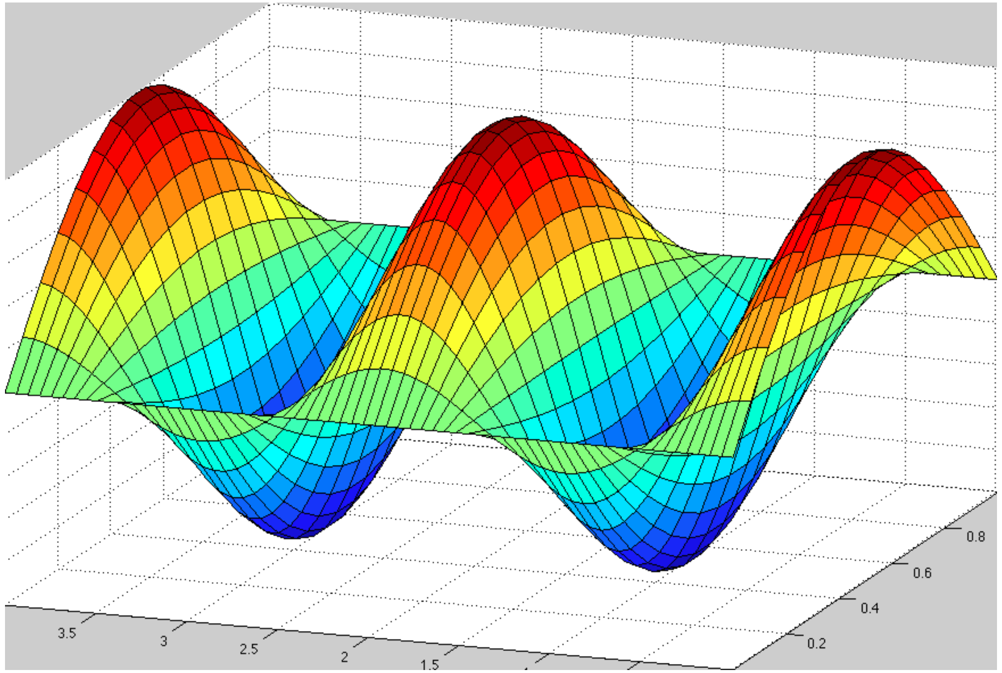
This course gives you the electromagnetic language behind many areas of electrical engineering.
Even if you don’t specialise in EM later, it explains *why* systems behave the way they do.

Concept of Field and Wave

Three-dimensional view

Two-dimensional view

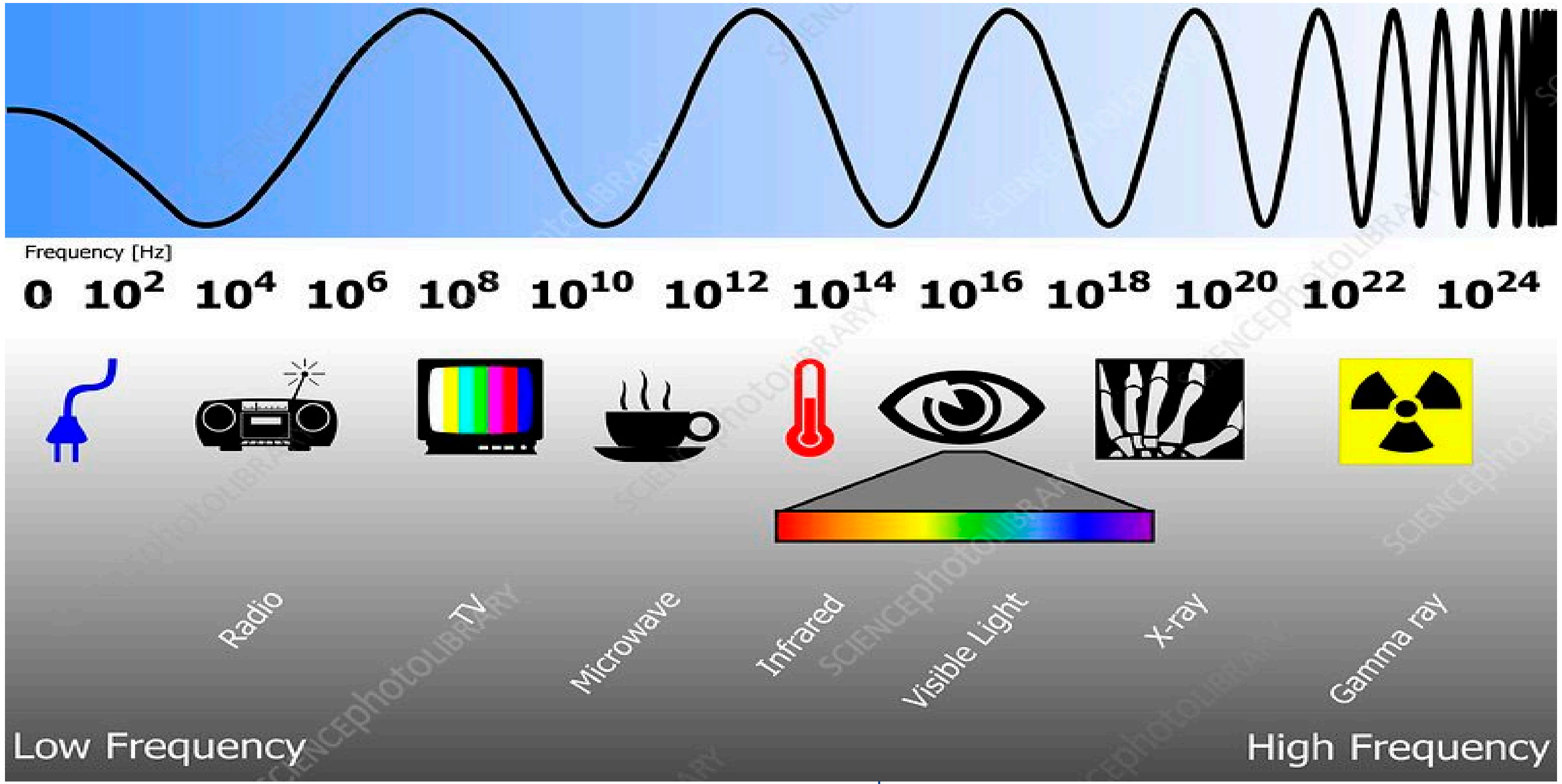




EM is both a field and a wave!

What do we mean by EM Field?

Why Part A and B ?



What is the wavelength of a 50 Hz electrical signal, and what would it be if the frequency increases to 500MHz?

$$\lambda = \frac{\text{velocity(m/s)}}{f\,(Hz)}$$

Part A (Wk 1-5)
Low frequency (i.e. 0-300Hz)
Large wavelength - Field*

Prof Rukmi Dutta

Part B (Wk 7-10)
High frequency (> 1MHz)
Small wavelength - Wave

Dr Dannielle Holms

Why Part A (low-frequency EM) matters for engineers

Electromagnetic fields are not just physics — they set engineering limits!

Electromagnetic fields determine:

- insulation stress and breakdown
- stored electrostatic energy & magnetic energy
- capacitance, inductance, and device size
- parasitic effects such as EMI

These limits exist **before** waves propagate.

Part A provides the physical constraints used in Part B.

In short:

You cannot design high-frequency systems without understanding EM fields.

You cannot design low-frequency systems and devices without understanding EM fields.

PART A Electromagnetic Fields (low-frequency EM)

Topic 1 - Static Electric Field and Capacitance (Week 1 and 2)

- **Workshop 1, 2, and Lab 1 (week 4/5)**

Topic 2 - Solutions of Electrostatic Problems (Week 3)

- **Workshop 3**

Topic 3 - Magnetic Field and Inductance (Week 4 and 5)

- **Workshop 4, 5, Lab 2, and 3 (week 6/7 & 9/10)**

Maxwell's equations

PART B
(HF EM)

-Feedback quiz – 5%

Mid-term Test on Week 7, Monday (30/03/2026) – 15%
-Lab starts from week 4 for even weeks and week 5 for odd weeks – 15%

- Assignment -
15%

-Final exam - 50% (part A: part B = 60%:40%)

Optional Bonus Marks for Participation

📘 **Lecture Participation**	🧩 **Workshop Participation**	ℹ️ **Key Rules** *(Details in Moodle)*
10 in-lecture questionnaires ⏱️ Open until **11:45 pm same day** ➕ **Up to +5 marks** (Mid-term)	**5 workshop activities** 🛠️ Problem-solving steps ➕ **Up to +5 marks** (Final exam)	⭐ Optional 🎓 Rewarding engagement 🧠 Your own understanding required 🚫 No mark above 100%
Progress bar: ▮▮▮▮▮▮▮▮▮▮ *("Each activity adds a little")*	**Progress bar:** ▮▮▮▮▮ *("Partial credit for each activity")*	📅 **Weeks 1 → 5** 📘 Lecture activities spread across lectures of Part A 🧩 Workshop activities every week
Participate when you can — each activity counts		

ELEC 3115

Lecture 1: Week 1, Monday

Topic 1- Static Electric Field and Capacitance

Outline

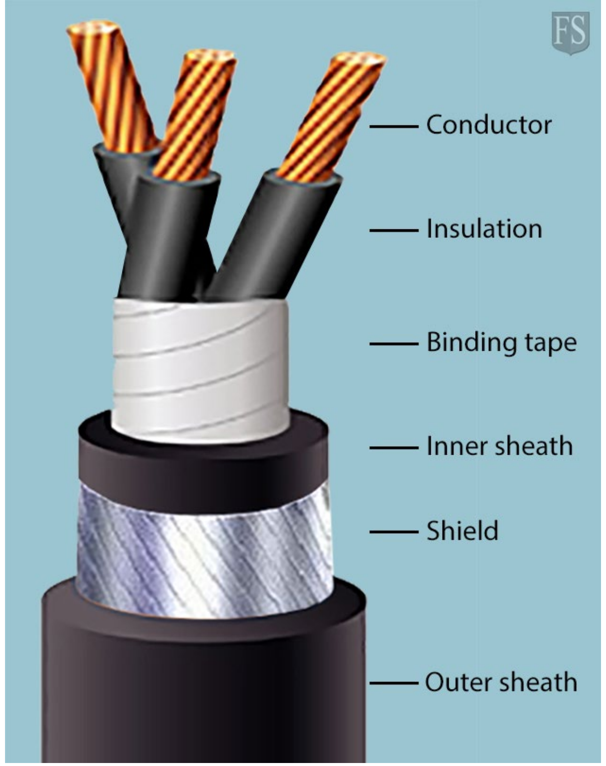
1.1 Finding of electric field and potentials

1.2 Electric field in material media

1.3 Capacitance formations

1.4 Electrostatic energy and force

Chapter 3 of your textbook by D.K Cheng



Is there an electric field here?

- Around the cable?
- Inside the metal?
- Inside the insulation?

If an electric field exists…

- Fields can be confined or leak
- Materials can respond differently
- Energy and force will also exist
- Energy can be stored in the field

Everything in this picture is governed in some way by the same static electric field.

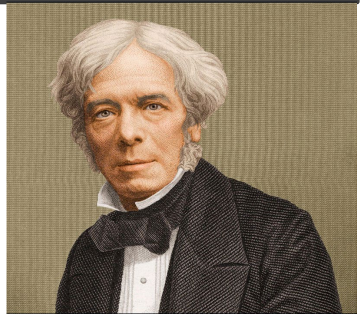
Today's learning goals

What you should be able to do after today,

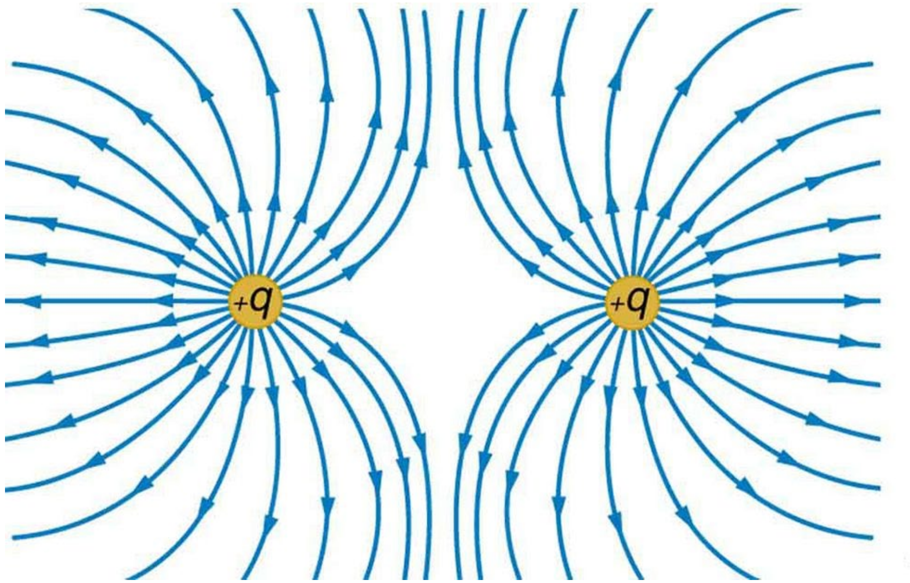
- Describe what an electric field and electric potential represent physically.
- Use Gauss's law to solve simple, symmetric electrostatic problems from real world.
- Explain how electrostatics connects to circuit laws (e.g. Kirchhoff's Voltage Law).

You are **not** expected to memorise derivations today — focus on *ideas, symmetry, and physical meaning*.

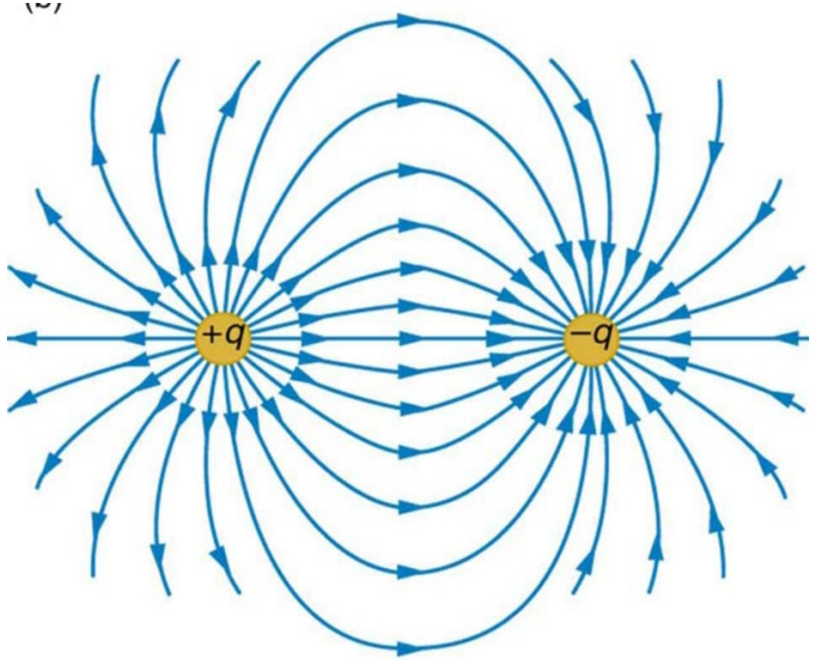
What is an electric field?



Michael Faraday



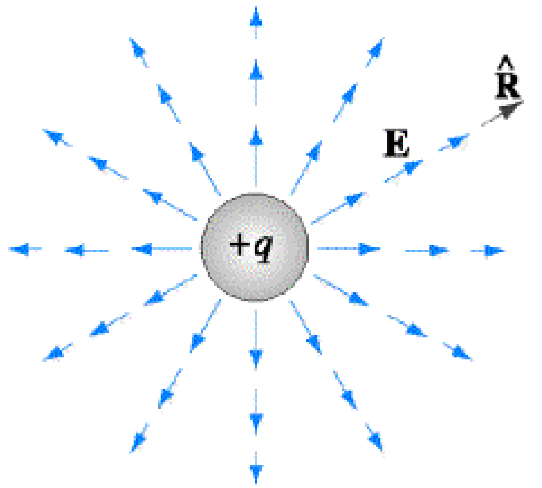
Repelling force



Attraction force

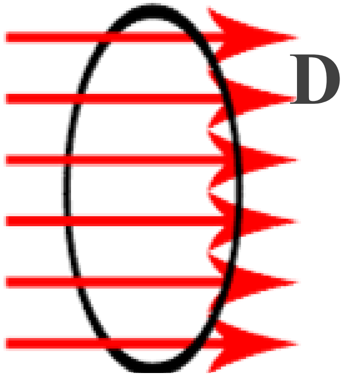
Charges experience force in the presence of other charges.

How do we measure electric field?



1. Electric field intensity $\vec{E}$: Force per unit stationary charge. SI unit N/C or V/m.

2. Electric flux density $\vec{D}$: Electric flux lines per unit area, SI unit C/m^2.



$$\vec{D} \propto \vec{E} \rightarrow \vec{D} = \varepsilon \vec{E}$$

ε : Permittivity of the medium

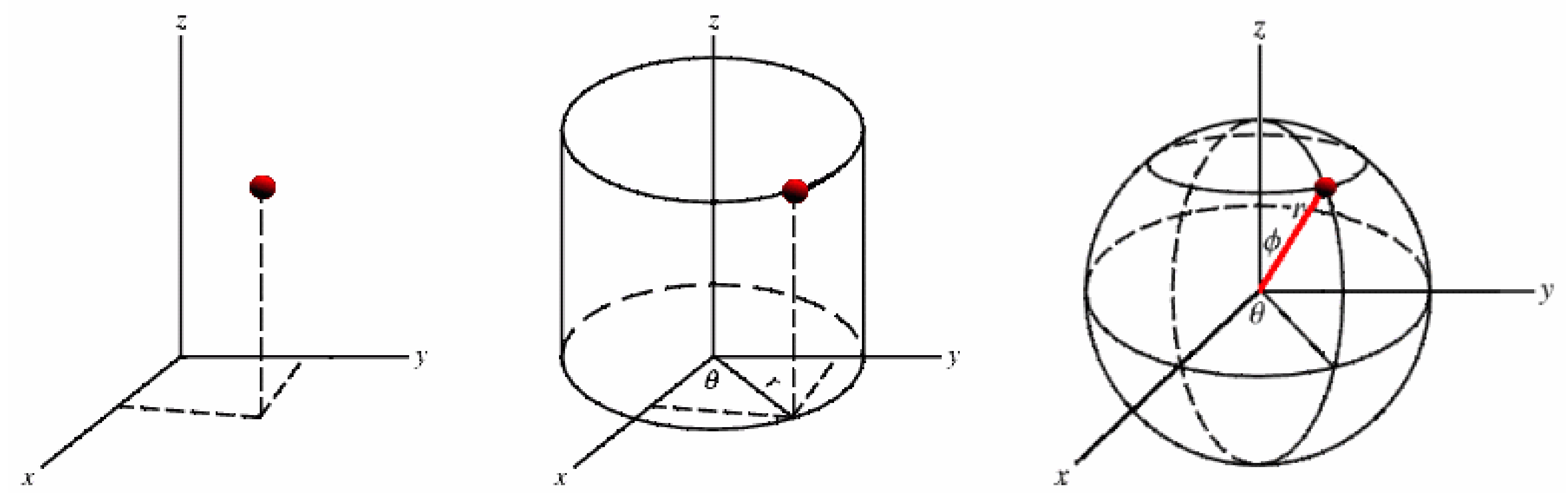
E and **D** are vectors and directions are same as the force **F** experienced by the charge.

Vector notations used in this course:

$$\vec{E} \text{ or } \mathbf{E} = |E| \vec{a}_R$$

How to represent field quantities E and D in space?

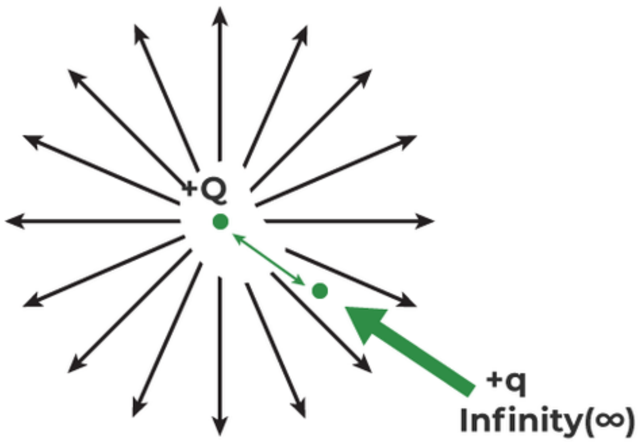
Point vectors in 3D space (defined by coordinate systems).

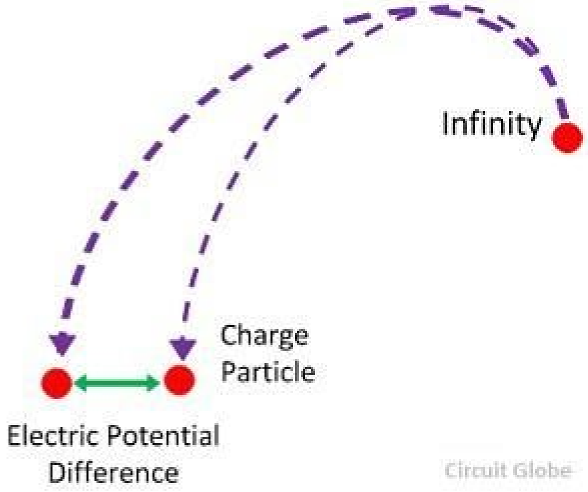


Cartesian or rectangular Cylindrical Spherical

Can you convert a vector of the cylindrical system to a vector of the Cartesian system?

What is an electric potential?



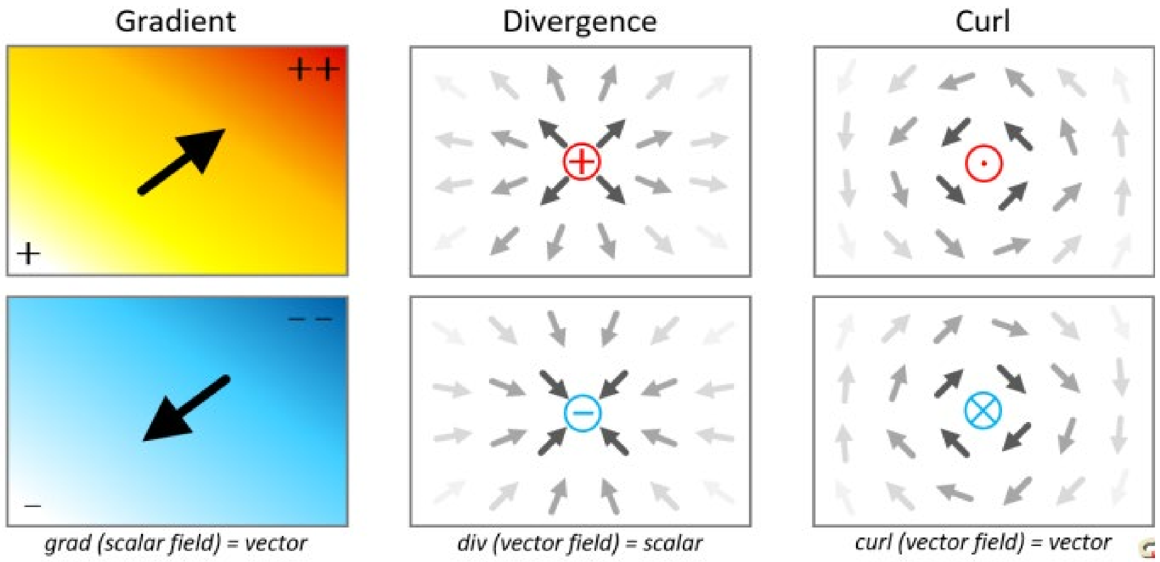


Work done to move a unit charge against an electric field is the electric potential.

Potential Difference $V_{ba} = V_b - V_a = \left(-\int_{\infty}^{b} \vec{E} \bullet d\vec{l} \right) - \left(-\int_{\infty}^{a} \vec{E} \bullet d\vec{l} \right) = -\int_{a}^{b} \vec{E} \bullet d\vec{l}$ Volt

V_{ba} is a 3-dimensional function, but scalar.
Negative sign in the definition means work done against the electric field **E**.

How to characterise vector E, D, and scalar *V (at a point in space)*



- A vector may diverge, converge, or pass through a point in space (Div), or it may circulate around a point (Curl).
- A scaler may increase or decrease in a particular direction – gradient (Grad)

We need Div (divergence), Curl, and Grad (gradient) operations of vector calculus to describe physical properties of a field at a point in space.

- **Div of E or D** → whether an electric charge exists that is the source or sink of the field
- **Curl of E or D** → whether field lines circulate
- **E = − (gradient of *V*)** → increase or decrease of potential V indicates where the related electric field exists.

Today: understand what these operators *tell us physically.* Later: you will use them mathematically.

grad, div, and curl operations with the Del operator

Gradient of a scalar function, say $V(x, y, z)$ → Partial differentiation!

$$\text{grad of V} = \left(\vec{a}_x \frac{\partial V}{\partial x} + \vec{a}_y \frac{\partial V}{\partial y} + \vec{a}_z \frac{\partial V}{\partial z} \right) = \left(\vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z} \right) V = \vec{\nabla} V$$

Del operator $\vec{\nabla} = \vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z}$ (Short-hand form to express partial derivatives in 3-dimensional coordinates!)

How do you write the Del operator in cylindrical coordinates (r, θ, z)?

Partially differentiating a vector: 2 ways

$$\vec{E} = \vec{a}_x E_x + \vec{a}_y E_y + \vec{a}_z E_z$$

div of $\vec{E} \rightarrow \vec{\nabla} \bullet \vec{\mathrm{E}}$

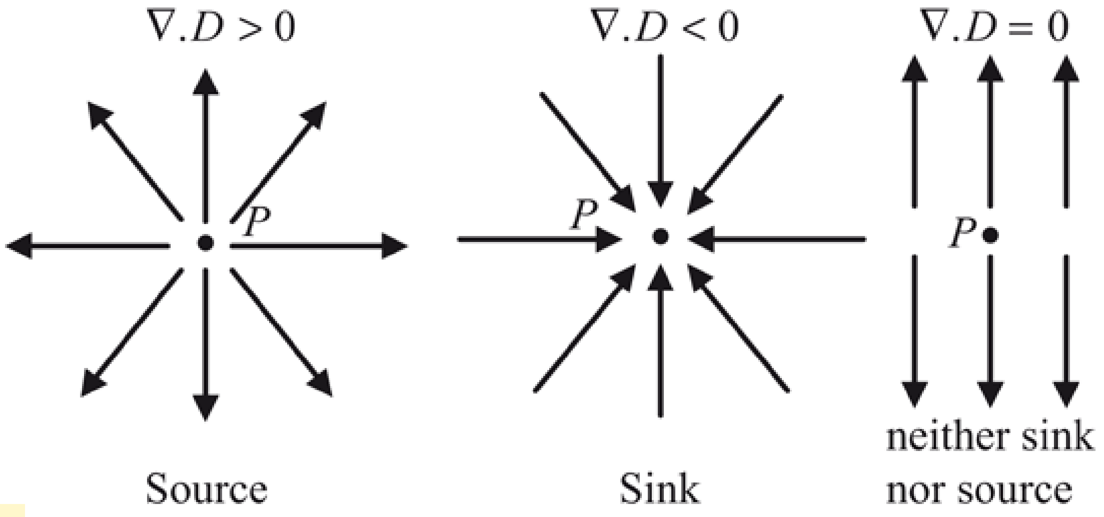
curl of $\vec{\mathrm{E}} \rightarrow \vec{\nabla} \times \vec{\mathrm{E}}$

$$\vec{\nabla} \bullet \vec{\mathrm{E}} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\vec{\nabla} \times \vec{\mathrm{E}} = \vec{a}_x \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right)$$

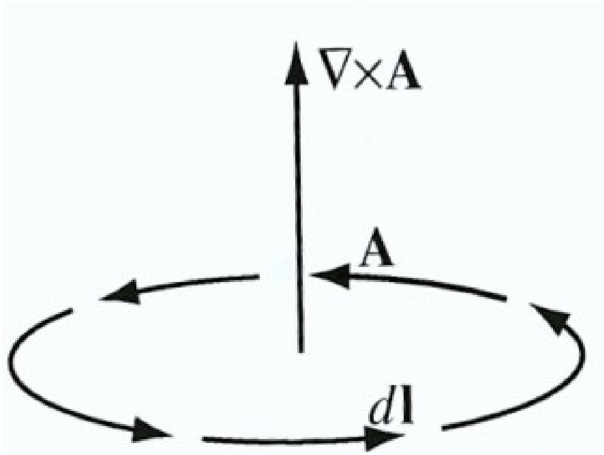
Div of a vector is a scalar and may have positive, negative or zero magnitude. A non-zero div measures the net outward/inward flux per unit volume enclosed by closed surfaces.

Divergence theorem: $\int_{vol} \left(\vec{\nabla} \bullet vector\right) dv = \oint_{s} vector \bullet d\vec{s}$



Curl of a vector is a vector and measures net circulation per unit area, and the direction is normal to the area.

Helmholtz's theorem: If, $\vec{\nabla} \times vector = 0 \rightarrow vector = \pm\vec{\nabla}(scalar)$



The gradient of a scalar is a vector.

Stokes theorem: $\int_{s} \left(\vec{\nabla} \times vector\right) \bullet d\vec{s} = \oint_{C} vector \bullet d\vec{l}$

We will need these theorems throughout the course for various derivations.

Pause & Reflect (Before we take a short break…)

✏️ **Take 1 minute to jot down:**

1. **One key idea** from this part of the lecture

2. **One question** that is still unclear

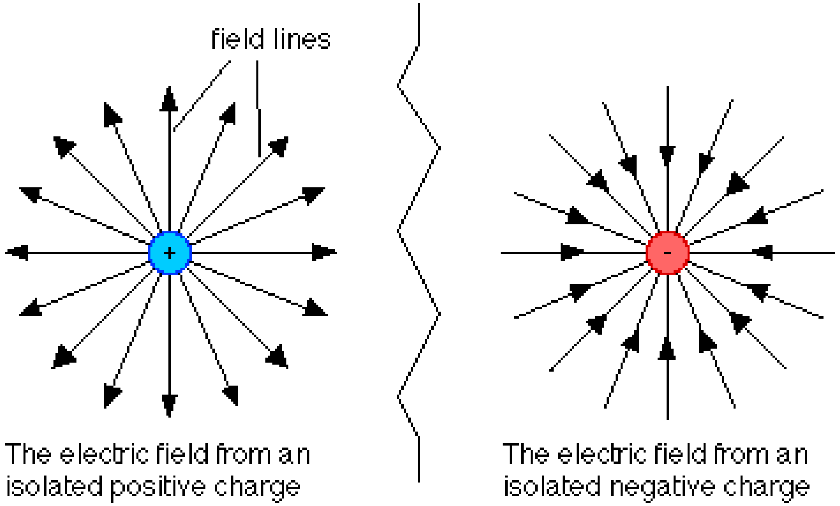
3. **One connection** to something you already know
   (another course, lab, real-world system, intuition)

💡 *You may be asked to submit one of these in bonus mark activities.*

Welcome back — anyone happy to share a key idea they jot down?

Div and Curl of a static electric field E

Let's observe the electric fields:



Does the electric field have a source or sink?

$$\vec{\nabla} \bullet \vec{D} \neq 0 \text{ or } \vec{\nabla} \bullet \vec{E} \neq 0$$

Is there circulation of flux lines?

$$\vec{\nabla} \times \vec{D} = 0 \quad \text{or} \quad \vec{\nabla} \times \vec{E} = 0;$$

$$\vec{E} = -\vec{\nabla} V$$

In electrostatics:

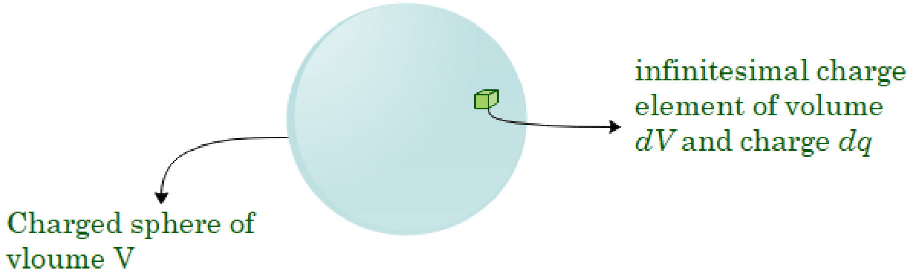
- field lines **start and end on charges → first postulate**
- field lines **do not circulate → second postulate**
  - **Electric field** is the negative **gradient of potential.**

When the source of an electric field is a charged body: charge density

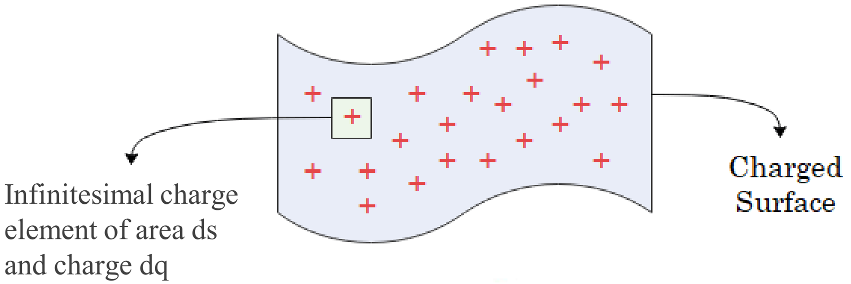
Charged body with a defined volume (e.g., sphere), or a surface (e.g., sphere's surface), or a long line (e.g., long conductor), we can define smooth average charge densities:

Volume charge density $\rho_v = \lim\limits_{\Delta v \to 0} \dfrac{\Delta q}{\Delta v}$ C/m^3

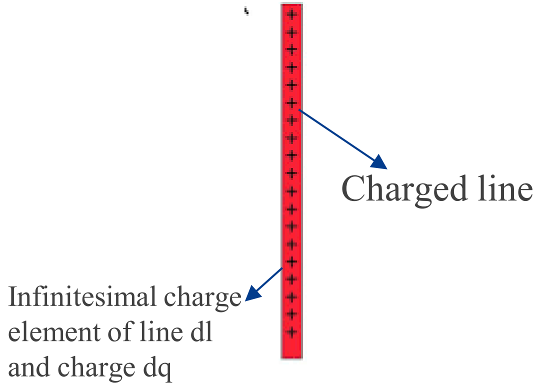
(Subscript *v* could be omitted.)



Surface charge density $\rho_s = \lim\limits_{\Delta s \to 0} \dfrac{\Delta q}{\Delta s}$ C/m^2



Line charge density $\rho_l = \lim\limits_{\Delta l \to 0} \dfrac{\Delta q}{\Delta l}$ C/m



Charge densities are point functions in the space coordinate.

Total charge $Q = \int_{vol} \rho dv$ or $\int_{s} \rho_s ds$ or $\int_{l} \rho_l dl$

Fundamental equations of Electrostatics

Electric fields in a medium diverge (or converge) from(or to) a source (or sink).

Postulate 1: $\vec{\nabla} \bullet \vec{D} = \rho \quad ; \quad \vec{D} = \varepsilon \vec{E} \quad \therefore \quad \vec{\nabla} \bullet \vec{E} = \dfrac{\rho}{\varepsilon}$

Permittivity of the medium (units: F/m):

$\varepsilon = \varepsilon_0 \varepsilon_r \quad$ [F/m]

$\varepsilon_o = 8.854 \times 10^{-12}$ F/m (permittivity of the free space).

ε_r= relative permittivity or the dielectric constant *(unitless)*.

Electric field does not circulate around a point.

Postulate 2: $\vec{\nabla} \times \vec{E} = 0,$ or $\vec{\nabla} \times \vec{D} = 0$

<u>What are the usages of the postulates?</u>

- Postulates are true at every point in the space of an electric field.
- ρ at a point can be found from the div of **E** or **D**.
- If the Curl of a point vector is zero, but div is non-zero, the vector could be **E** and **D**.
- Since curl is zero, **E** can be found from the gradient of scalar potential V.

How to use the postulates to solve any engineering problem

(Integral form of postulates)

Taking volume integral of postulate 1, $\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\varepsilon}$

$$\int_{vol} \left(\vec{\nabla} \bullet \vec{E}\right) dv = \frac{1}{\varepsilon} \int_{vol} \rho dv = \frac{Q_{enclosed}}{\varepsilon},$$

$Q_{enclosed}$: total charge enclosed within the closed surface (volume).

Apply divergence theorem to LHS, $\int_{vol} \left(\vec{\nabla} \bullet \vec{E}\right) dv = \oint_s \vec{E} \bullet d\vec{s}$

Substituting,

$$\therefore \oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon} \quad \text{or} \quad \oint_s \vec{D} \bullet d\vec{s} = Q_{enclosed}$$

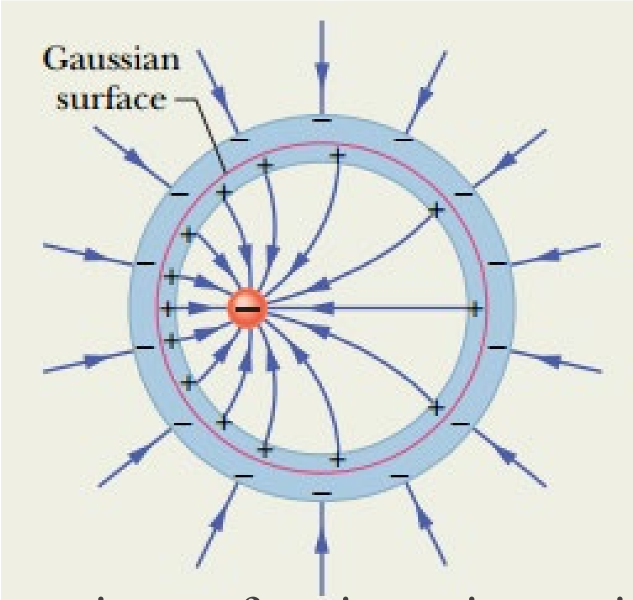
(Integral form of the first postulate)

Does it look familiar? **Gauss's law**

Note that, $Q_{enclosed} = \int_{vol} \rho dv \text{ or } \int_s \rho_s ds \text{ or} \int_l \rho_l dl$

Gauss's Law and Gaussian surface

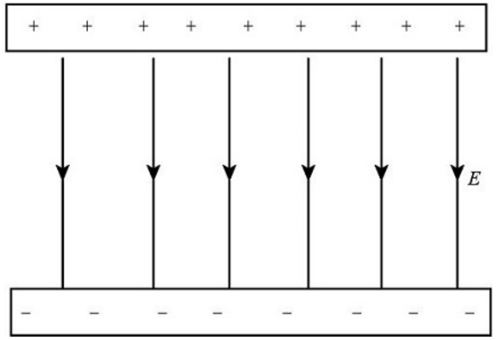
$$\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon}$$



Gaussian surface is equipotential

The surface integral becomes simple when surfaces (called **Gaussian surfaces**) can be found on which $\vec{E}$ remains constant (i.e., has symmetry) and the direction of E is normal to the surface.

Let's consider a uniform electric field between two large parallel plates of capacitors. Which of the following shapes could give us a Gaussian surface to calculate the electric flux through it?



(a) A closed sphere with the source charges at its centre.
(b) A cylinder with its axis parallel to the electric field lines.
(c) A cube with one face perpendicular to the electric field lines.
(d) A cone with its vertex pointing towards the source charges.

Integral form of the second Postulate $\vec{\nabla} \times \vec{E} = 0$

Taking surface integral of postulate 2 and then applying Stokes's theorem, we get the integral from of the second postulate. $\int_s \left(\vec{\nabla} \times \vec{E}\right) \bullet d\vec{s} = 0 \rightarrow \oint_C \vec{E} \bullet d\vec{l} = 0$

Substituting $\vec{E} = -\vec{\nabla} V$

$$\oint_C \left(-\vec{\nabla} V\right) \bullet d\vec{l} = \oint_c dV = 0 \qquad \because \vec{\nabla} V \bullet \vec{a}_l = \frac{dV}{dl}$$

The sum of the electrical potential differences (voltages) around any closed path (i.e. circuit) is zero. Does this sound familiar? What is the common circuit theory law for voltage?

Kirchhoff's voltage law (KVL).

Potential difference between two arbitrary points:

$$V_{ba} = -\int_a^b \vec{E} \bullet d\vec{l} = -\int_a^b \left(-\vec{\nabla} V\right) \bullet d\vec{l} = \int_a^b dV = V_b - V_a$$

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How to apply Gauss's Law (problem-solving strategy)

Before writing any equations:

1. **Identify symmetry** (spherical, cylindrical, planar).
2. **Choose a Gaussian surface** that matches the symmetry.
3. Decide whether **E is constant or zero** on that surface.
4. Apply,

$$\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon} \quad \text{or} \quad \oint_{\mathrm{s}} \vec{D} \bullet d\vec{s} = Q_{enclosed}$$

Gauss's law is powerful **only when symmetry is used correctly.**

Example 1

A scientist studying storm prediction needs to find the electric field **E** inside and outside a charged cloud, and the variation of **E** at a radial distance R from the centre of the cloud. How can it be done?

Solution steps:

1. Identify symmetry
2. Choose a Gaussian surface
3. Apply Gauss's law,

$$\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon_0} = \frac{\rho \int_{vol} dv}{\varepsilon_0}$$

4. Find $\mathbf{E}_{\text{inside}}$ at an arbitrary radius $0 < R < b$

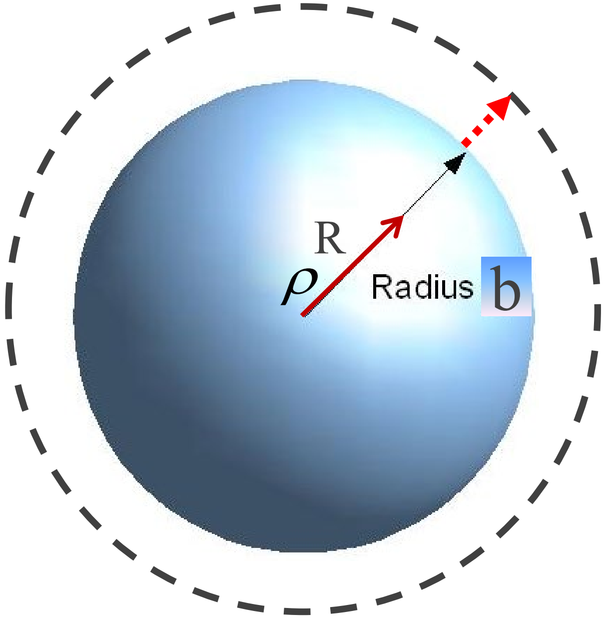
$$\left|E 4\pi R^2\right| = \frac{\rho}{\varepsilon_0}\left(\frac{4}{3}\pi R^3\right) \rightarrow |E| = \frac{\rho R}{3\varepsilon_0}$$

Radius of the sphere: b

Volume charge density : ρ

E at any arbitrary radial distance R inside the sphere:

$$\vec{E}_{inside} = \left|\frac{\rho R}{3\varepsilon_0}\right| \vec{a}_R \ \text{V/m}, \quad \text{for } \ 0 < R \le b$$



5. Define the Gaussian surface to find **E** at any radial distance R outside.

Note that the total charge of the sphere is enclosed by the chosen surface (dotted spherical surface).

$$\oint_s \vec{E}_{outside} \bullet d\vec{s} = \frac{\rho \int_{vol} dv}{\varepsilon_0} \rightarrow E_{outside}\left(4\pi R^2\right) = \frac{Q_{sphere}}{\varepsilon_0} = \left|\frac{\rho \frac{4}{3}\pi b^3}{4\pi\varepsilon_0 R^2}\right|, R > b$$

$$\left|E_{outside}\right| \vec{a}_R = \left|\frac{\rho b^3}{3\varepsilon_0 R^2}\right| \vec{a}_R, \quad R > b$$

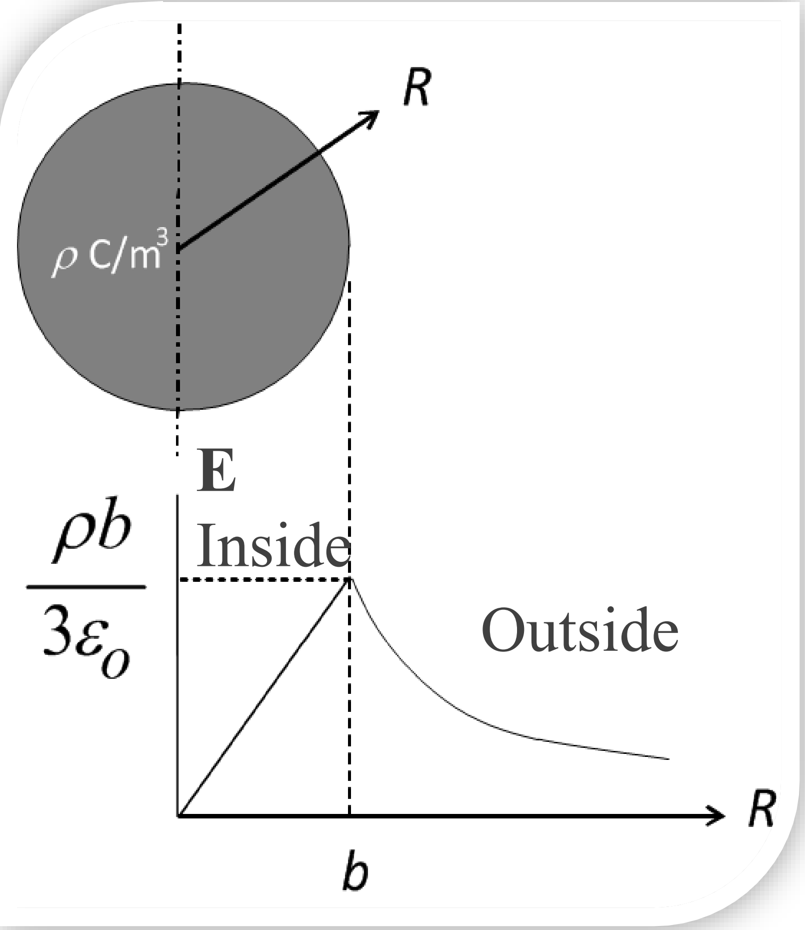
$$\vec{E}_{inside} = \left|\frac{\rho R}{3\varepsilon_o}\right| \vec{a}_R \text{ V/m for } 0 < R \leq b$$

$$\vec{E}_{outside} = \vec{a}_R \frac{\rho b^3}{3\varepsilon_o R^2} \text{ V/m for } R > b$$

Physical interpretation

- Electric field **increases linearly** with distance from the centre.
- Field is **zero at the centre** ($R = 0$).
- Maximum value occurs at the **surface** ($R = b$).
- This behaviour is a direct consequence of **spherical symmetry**.

Variation of **E** with **R**



Additional practice question

Where does the electric field reach the maximum value? If E_{max} = 3MV/m and b = 10m, what is the value of charge density in the cloud?

Given, $\varepsilon_o = 8.854 \times 10^{-12}$ F/m

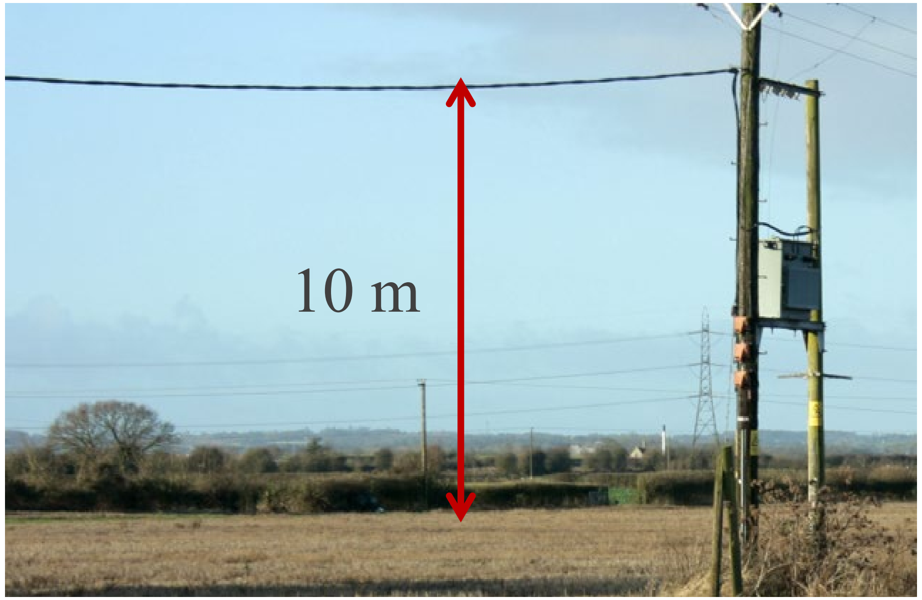
Ans: 796.86 μC/m^3

Example 2

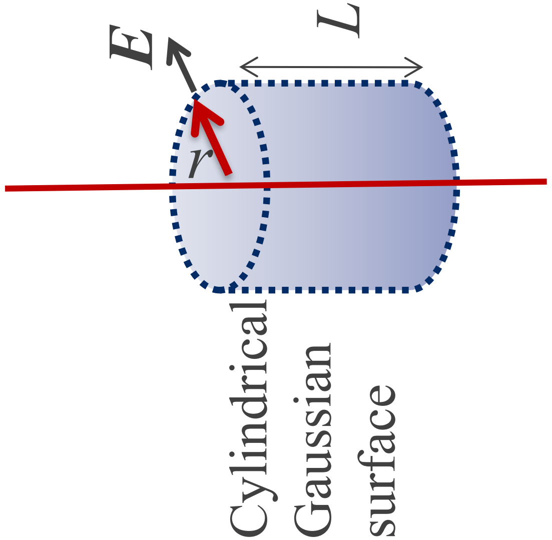
Consider a very long transmission line located above the ground at a height where any influence of the ground is negligible. The line is charged with a uniform line charge density ρ_l.

(i) Find an expression of electric field **E** everywhere in the space around the transmission line

(ii) Find the magnitude of the electric field intensity at ground level directly below the line when the height from the ground to the line is 10m, and the uniform line charge density ρ_l is 10^{-7} C/m.



Solution



1. Identify symmetry

2. What should be the Gaussian surface here?

3. Apply $\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon}$

4. What is the $Q_{enclosed}$ in the Gaussian surface ?

$$Q_{enclosed} = \int_0^L \rho_l dl = \rho_l L$$

5. What is Gaussian surface area? $\left|\oint_s d\vec{s}\right| = 2\pi r L$

6. Substitute in GL and solve for E,

$$\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon_0} \rightarrow 2\pi r L |E| \vec{a}_R = \frac{\rho_l L}{\varepsilon_0}$$

$$\therefore \vec{E} = \left|\frac{\rho_l}{2\pi\varepsilon_0 r}\right| \vec{a}_r \quad \text{V/m}$$

(ii) Find the magnitude of the electric field intensity at ground level directly below the line when height from the ground to the line is 10m and the uniform line charge density ρ_l is 10^{-7} C/m.

$$\varepsilon_o = 8.854 \times 10^{-12} \text{ F/m}$$

Ans: 179.755 N/C or V/m

$$\vec{E} = \left| \frac{\rho_l}{2\pi\varepsilon_0 r} \right| \vec{a}_r \quad \text{V/m}$$

At a point directly below the line, distance from line $r = h = 10$m

Given, $\rho_l = 10^{-7}\ C/m$

Physical interpretation:

- Cylindrical symmetry is the key — without it, Gauss's law would not simplify this problem.
- Electric field **decreases as** $1/r$
- Field is strongest **closest to the conductor**
- This behaviour will explain later:
  - insulation design limits
  - corona onset in transmission lines

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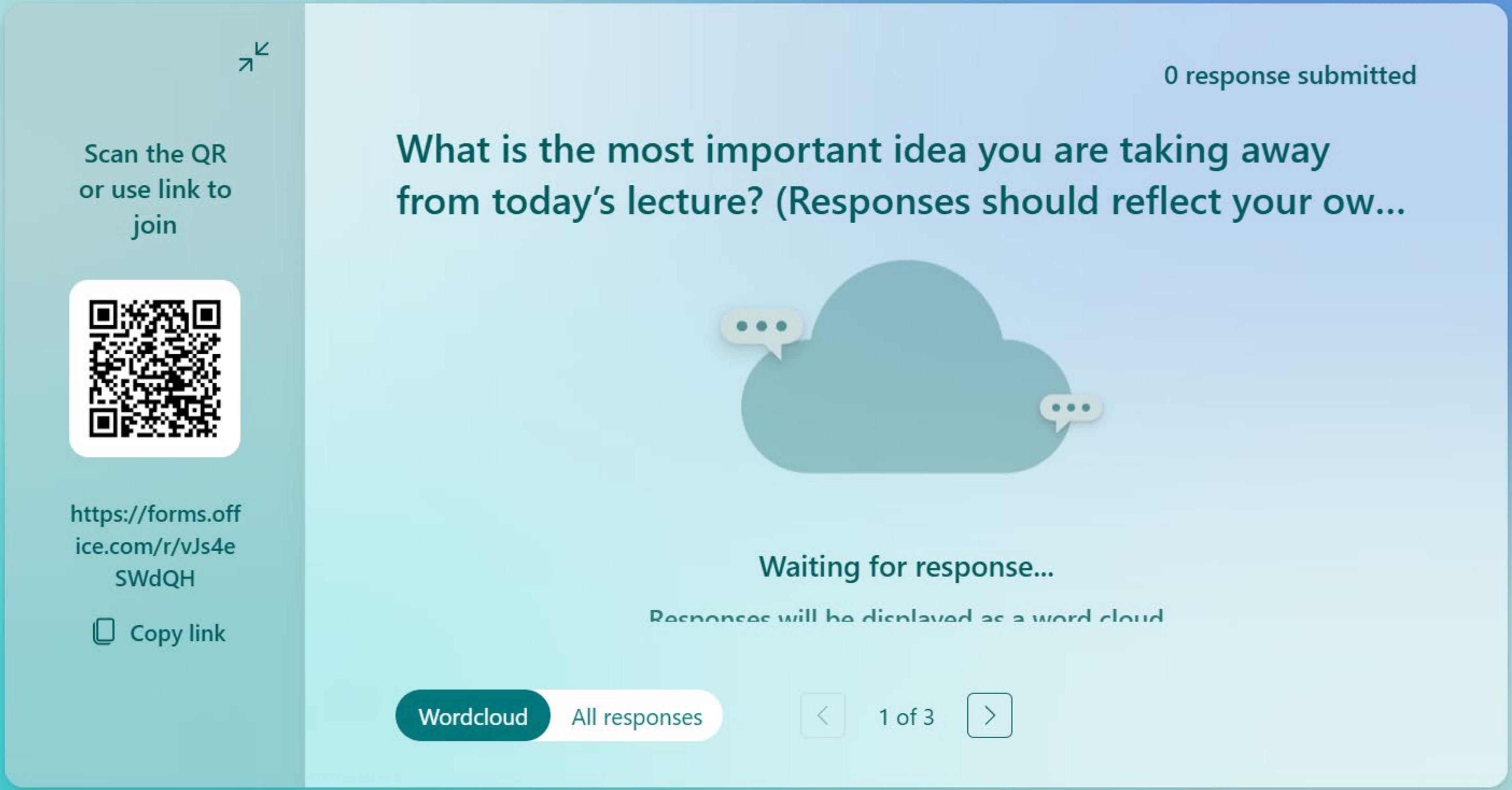
Before You Go… (1 minute reflection)

This reflection will count toward optional bonus marks.



Key takeaways from today

Electric fields describe how charges influence space.

Gauss's law works best when **symmetry is present.**

Electrostatic behaviour leads directly to:

- Gauss's law (first postulate)
- Kirchhoff's Voltage Law (← second postulate)

Before Lecture 2 and Workshop 1:
Review Gauss's law examples and focus on *choosing the Gaussian surface*.

1. If the surface charge density of a large sheet is ρ_s C/m^2 . Show that **E** at any arbitrary point of the free space above the sheet is given as,

$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_\mathrm{n}$$

Where $\boldsymbol{a}_n$ *is unit vector in the direction normal to the surface of the sheet. (hint: Example 3-6)*

2. Show that potential difference at an arbitrary radial distance R from a discrete charge q with respect to a reference at infinite is given as,

$$V = \frac{q}{4\pi\varepsilon R} \quad \mathrm{V}$$

Using this, can you explain why Gaussian surface around q is at equipotential.

Further practice (Recommended)

- Practice Chapter-2 examples of the textbook if you want to strengthen your skills of vector analysis.
- Practice Example 3-1, 3-2, 3-4 from the textbook by D.K Cheng.